BUILD - TEST - EXPLAIN

Arduino Foundations Workbook

Two complete labs: digital output and servo control

LEVEL	FORMAT	EVIDENCE
Beginner	**2 guided labs**	**Code, wiring, checks and quiz**

This workbook supports the complete REES52 Academy course. Watch each linked lesson in the Academy, build with power disconnected, then record your evidence here.

Learner name: ______________________________ Date: _____________________

Before you build

Materials checklist

ITEM	QTY	CHECK
Arduino Uno compatible board	1	[ ]
LED	1	[ ]
220 ohm resistor	1	[ ]
SG90 servo	1	[ ]
Breadboard	1	[ ]
Jumper wires	6	[ ]

Safety and setup

- Disconnect USB power before changing connections.
- Always use a current-limiting resistor with the LED.
- Keep fingers and wires clear of the moving servo horn.

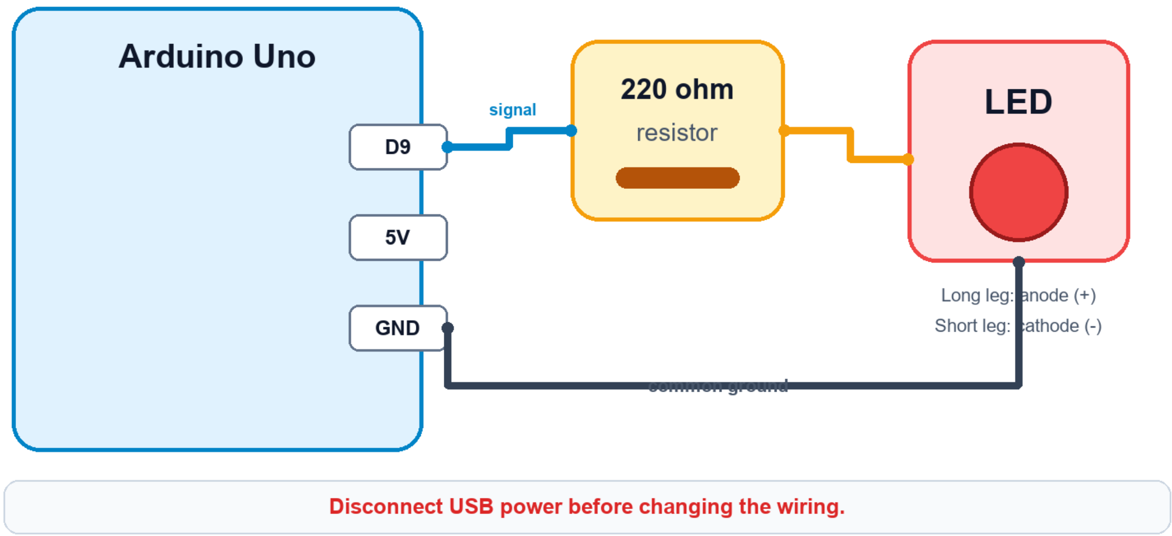
Evidence standard

A completed lab includes: a clear wiring photo, code that compiles, one observed result, one debugging note, and a short explanation in your own words.

Wiring reference

Arduino LED Wiring

Digital output practice circuit



Verify every label against your board before applying power. Board variants can use different printed pin names.

Pre-power check

[ ] Supply voltage checked [ ] Common ground connected [ ] No exposed wire strands [ ] Polarity confirmed

Lab 1: Blink an LED

Use a digital output to create a repeatable on/off pattern, then change the timing and explain the result.

Build sequence

1. Connect the LED through a 220 ohm resistor to digital pin 9.
2. Connect the LED cathode to GND and inspect polarity.
3. Upload the sketch and observe three complete blink cycles.
4. Change both delay values and record the new pattern.

Starter code

```
const int ledPin = 9;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(ledPin, HIGH);
  delay(500);
  digitalWrite(ledPin, LOW);
  delay(500);
}
```

Observation

What happened?

____

What did you change?

____

Explain why it worked:

____

Lab 2: Control a Servo

Generate position commands for a servo and compare two target angles without forcing the horn against a mechanical stop.

Build sequence

1. Connect signal to D6, power to 5 V and ground to GND.
2. Upload the sweep-between-two-angles sketch.
3. Confirm the servo reaches both target positions smoothly.
4. Reduce the travel range if the servo chatters or stalls.

Starter code

```
#include <Servo.h>

Servo arm;

void setup() {
  arm.attach(6);
}

void loop() {
  arm.write(20);
  delay(800);
  arm.write(100);
  delay(800);
}
```

Observation

What happened?

____

What did you change?

____

Explain why it worked:

____

Troubleshooting map

SYMPTOM	CHECK FIRST	NEXT TEST
No output	Power, board and selected port	Run the smallest known-good sketch
Unstable readings	Loose ground or long signal wires	Print raw values and average readings
Wrong direction	Motor or output mapping	Test one output channel at a time
Upload fails	USB cable and selected board	Close Serial Monitor and reconnect

Debugging record

Issue found: ____

Test performed: ____

Final fix: ____

Course check

Choose one answer for each question. Submit the online quiz in REES52 Academy for scoring and feedback.

1. Why is a resistor used with the LED?

A) To limit current B) To store code C) To reverse polarity D) To increase voltage

2. Which function configures an Arduino pin?

A) pinMode() B) delay() C) attach() D) print()

3. What unit does delay() use?

A) Milliseconds B) Volts C) Degrees D) Bytes

4. Which library controls a hobby servo?

A) Servo B) WiFi C) DHT D) Stepper only

5. What should you do before rewiring?

A) Disconnect power B) Increase voltage C) Remove the resistor D) Short 5 V to GND

Reflection

The concept I can now explain clearly is:

____

My next improvement will be:

____

Completion evidence

Use this final page before marking the course complete.

EVIDENCE	READY
All lesson videos watched	[ ]
Wiring built and photographed	[ ]
Starter code modified and saved	[ ]
Measured result recorded	[ ]
Online quiz passed	[ ]
Project explanation written in learner's own words	[ ]

Video source

The course lessons use tutorials published by REES52 at https://rees52.com/pages/tutorials. Links are also available inside each Academy lesson.

Course completion certificates show learning progress inside REES52 Academy. They are not a degree, licence, or government-accredited qualification.